TECHNICAL NOTE / 01

SLAC circuit control optimization and Toffoli-count derivation

Implementation note for the complete signed inverse-square LCU, its multiscale transform, and its two-sided wavelet preconditioner. Prepared 14 October 2026.

Main results for slides

Let $\mathbf{N = 2^n}$, n ≥ 3, and choose $\mathbf{n_{ref} = 2n + 4}$. Under the coherent-Toffoli convention below, the implemented circuits have:

$$T_{\mathrm{LCU}}(n) = 100n^2 + 142n - 32$$

$$T_{\mathrm{MS}}(n) = \frac{4n^3 + 327n^2 + 371n - 126}{3}$$

$$T_{\mathrm{PMS}}(n) = \frac{4n^3 + 327n^2 + 383n - 150}{3}$$

These expressions are derived from the gate structure, not obtained by polynomial fitting. They match the Qualtran call-graph counts for every tested n = 3, ..., 16.

What operator and circuit are counted?

The base circuit encodes the paper's **truncated signed $1/j^2$ operator**, with diagonal $-\pi^2/3$, paired shift coefficients $2(-1)^{1+j}/j^2$ for $1 \le j < N/2$, and zero midpoint-shift coefficient. A finite reference register approximates these weights through an integer inequality test.

Base: PREP(steps 1-6) → SIGN → COPY → SELECT → PREP†.
Multiscale: wrap the base operator by W and W†.
Preconditioned: additionally place P on both sides, using independent signal ancillas.

Counting convention

Each And and And† is counted as one **coherent Toffoli**. All compute/uncompute calls are included. Logical rotations, Clifford gates and explicit T gates are separate categories. These are counts for the selected implementation, **not a proof of globally minimal gate counts**.

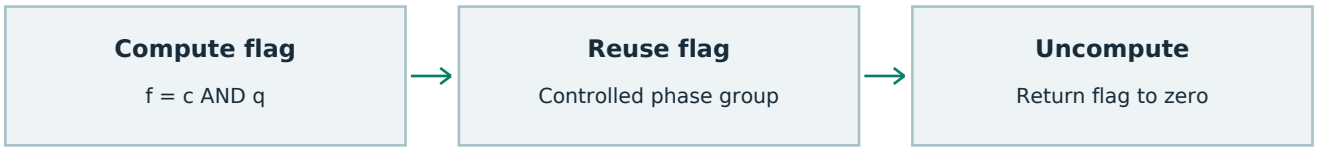
Excluded: rotation synthesis, a nullspace-projector circuit, QLSA repetitions/amplitude amplification, and physical error correction.

TECHNICAL NOTE / 02

How the control overhead is reduced

1. Reuse the control conjunction inside a QFT

An externally controlled QFT has groups of phase gates sharing the condition that the external enable bit c and a QFT bit q are both 1. Compute their conjunction into a clean flag once, use that flag throughout the phase group, and then uncompute it. The control bits are not changed during this group.



One Toffoli to compute + one to uncompute; phase rotations are counted separately.

This avoids separately rebuilding the same conjunction for every phase in the group. It costs two extra Toffolis per nonempty phase group, plus the cost of the singly controlled phase gates themselves. The latter costs are **not removed** from the ledger.

In **SharedControlQFT**, these phases are represented by a controlled PhaseGradientUnitary. Controlled Hadamards are lowered to rotations and CZ; controlled swaps cost one Toffoli plus two CNOTs each.

2. Reuse nested zero-prefix controls in the multiscale recursion

The successive infrared branches need conditions such as $b_0 = 0$, $b_0 b_1 = 00$, and so on. Instead of attaching all earlier controls to every elementary gate, build a ladder of reusable flags:

$$f_1 = \neg b_0, \qquad f_{s+1} = f_s \wedge (\neg b_s)$$

Each deeper QSWT gets a single enable flag. The branch bits are produced sequentially by the QSWTs; after the recursion, the flag ladder is reversed and cleaned. This is a shared-prefix idea related to unary iteration, not a call to the generic Qualtran unary-iteration SELECT routine.

One W uses one uncontrolled n-qubit step and controlled steps of widths n-1 down to 2. The first prefix is obtained using X gates; n-3 additional prefixes each need one compute/uncompute pair. Thus the prefix overhead is **2(n-3) per W**, or **4(n-3) for W and its adjoint**.

What remains unchanged

The construction still makes **2(n-1) one-scale QSWT calls** across W and $W^\dagger$. No transform level is omitted. The principal saving relative to the previous recursively controlled implementation comes from sharing QFT phase-group controls; prefix sharing alone is not credited with a separate saving when Qualtran already shares those prefixes.

TECHNICAL NOTE / 03

Deriving the multiscale cost

One k-qubit QSWT step

For k ≥ 2, the following table follows the actual component decomposition. The QFT figures include its chosen controlled-phase implementation; they are not universal lower bounds for a QFT.

Component	Uncontrolled Toffolis	Shared-control Toffolis
QFT on k qubits	k(k-1)	k(k-1) + 2(k-1) + floor(k/2)
Inverse QFT on k-1 qubits	(k-1)(k-2)	(k-1)(k-2) + 2(k-2) + floor((k-1)/2)
Shift	0	0
Mix (two pattern predicates)	4(k-2)	4(k-1)
Split	0	1
Block phase/permutation	2(k-2)	2(k-1)

The QFT contains k(k-1)/2 controlled phase gates, each contributing two coherent And/And-adjoint Toffolis in this decomposition. Extra external control introduces k-1 shared predicate pairs and floor(k/2) controlled swaps. Each pattern-predicate ladder costs two Toffolis per internal conjunction.

$$t(k) = k(k-1) + (k-1)(k-2) + 4(k-2) + 2(k-2) = 2k^2 + 2k - 10$$

The extra cost of the two QFTs is 5k-7, using floor(k/2) + floor((k-1)/2) = k-1. Mix, split and block add another 4 + 1 + 2 = 7. Hence:

$$t_c(k) = t(k) + 5k$$

Sum over the scales, including both transform directions

$$T_{W,\,W^\dagger} = 2t(n) + 2\sum_{k=2}^{n-1}[t(k) + 5k] + 4(n-3)$$

Substituting t(k) = $2k^2$ + 2k - 10 gives the summand $2k^2$ + 7k - 10. Using the standard sums of k and k^2 from 2 through n-1 yields:

$$T_{W,\,W^\dagger} = \frac{4n^3 + 27n^2 - 55n - 30}{3}$$

At n = 16: one-step costs and all prefixes sum to **7,462 Toffolis** for W and $W^\dagger$ together. The formulas above were also checked against the component call graphs at k = 2, ..., 8.

TECHNICAL NOTE / 04

Deriving the complete LCU cost

Write **q = n-1**, **r = n_{ref}**, and **L = 2q+r**. The predicate is m $j^2 \geq M2^{2\mu}$ for failure, with $M = 2^r$. Its result is copied into the failure flag only on the nonzero-coefficient branch.

Executable arithmetic from controlled adders

A controlled b-bit addend loaded into an L-bit accumulator uses b And/And-adjoint pairs plus L-1 adder pairs: **2b + 2(L-1)** Toffolis. A schoolbook product adds b shifted by each bit of an a-bit multiplicand:

$$T_{\text{prod}}(a,b) = \sum_{i=0}^{a-1} [2b + 2(a+b-i-1)] = a(4b+a-1)$$

Operation	Formula	n=16, r=36
Square j using two q-bit copies	S = q(5q-1)	1,110
Product m times j squared	R = r(8q+r-1)	5,580
One equal-width comparator	C = 4(2q+r)-2	262

The comparator uses L-1 mixer compute/uncompute pairs, each costing four Toffolis, and a final single-bit comparison pair costing two: C = 4(L-1)+2. Copying j for the square and wiring the one-hot left-hand side use no Toffolis.

Clean predicate and complete LCU

Compute the square and product; compute the comparison; use one Toffoli to copy the comparison conditioned on branch a; uncompute comparison, product and square. Therefore:

$$T_{\text{test}} = 2S + 2R + 2C + 1$$

PREP and PREP† each contain this clean test. SELECT costs 2q Toffolis. SIGN is XZX = -Z on the least significant j bit; COPY is a CNOT into a marker that is deliberately not uncomputed. They add no Toffolis. Controlled-H gates and preparation rotations are accounted for separately.

T_{LCU} = 2 T_{test} + 2q = 4S + 4R + 4C + 2 + 2q. Expanding gives:

$$T_{\text{LCU}}(n,r) = 20q^2 + 32qr + 4r^2 + 30q + 12r - 6, \qquad q = n-1$$

For the precision schedule r = 2n+4, substitute q = n-1:

$$T_{\text{LCU}}(n) = 100n^2 + 142n - 32$$

At n=16: T_{test} = 2(1,110) + 2(5,580) + 2(262) + 1 = 13,905. Thus T_{LCU} = 2(13,905) + 30 = **27,840**.

TECHNICAL NOTE / 05

Preconditioner and combined formulas

Scale-controlled rotations for P

The Laplacian preconditioner is P = diag(I_2, $2^{-1}I_2$, $2^{-2}I_4$, ...). On scale s its signal ancilla receives H $R_z(-2\theta_s)$ H, whose zero-to-zero amplitude is $\cos(\theta_s) = 2^{-s}$.

$$\theta_s = \arccos(2^{-s}), \qquad \delta_\ell = \theta_{n-1-\ell} - \theta_{n-\ell}$$

Start with the unconditional phase θ_{n-1}. Each zero-prefix of length ℓ then controls the phase increment δ_ℓ. These increments telescope to the correct scale angle. Using full scale angles on overlapping prefixes would be incorrect.

One P computes n-2 prefix flags and uncomputes them. Each controlled R_z decomposes into two rotations and two CNOTs, with no additional Toffolis. The two P copies use independent signal ancillas so their projected blocks multiply correctly.

$$T_P = 2(n-2), \quad T_{P,P} = 4(n-2), \quad R_P = 2n-1$$

Per P, the other gates are 2(n-1) CNOTs, 2(n-1) X gates and two H gates. There are n-2 clean prefix workspace qubits. The P normalization is 1.

Add the three components

$$T_{\mathrm{MS}} = T_{\mathrm{LCU}} + T_{W,W^\dagger}, \qquad T_{\mathrm{PMS}} = T_{\mathrm{MS}} + 4(n-2)$$

$$T_{\mathrm{MS}}(n) = \frac{4n^3 + 327n^2 + 371n - 126}{3}$$

$$T_{\mathrm{PMS}}(n) = \frac{4n^3 + 327n^2 + 383n - 150}{3}$$

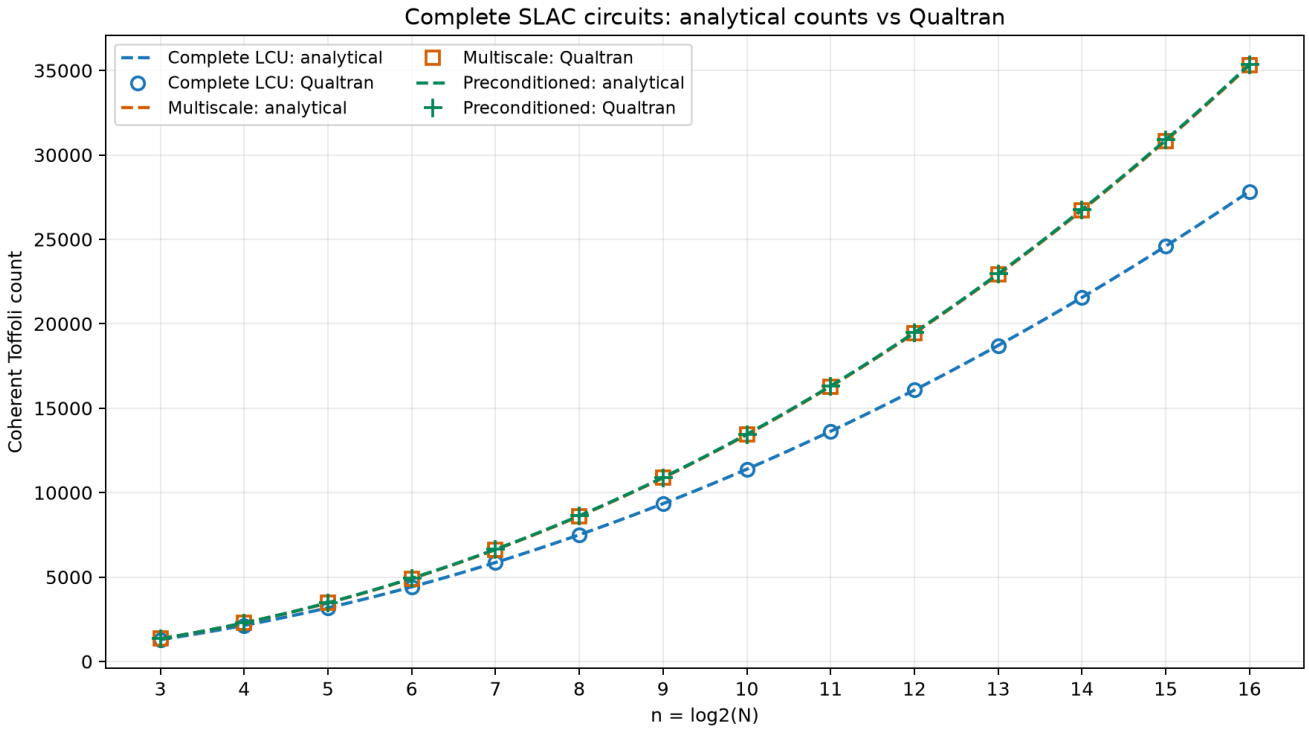
Component at n=16, r=36	Toffoli contribution
Complete base LCU	27,840
W and W-adjoint	7,462
Both P circuits	56
Multiscale total	35,302
Preconditioned total	**35,358**

Only the first three rows are additive; the last two are subtotals. These formulas include every clean arithmetic step in the implemented block encoding, but do not include a separate projection away from the constant mode.

TECHNICAL NOTE / 06

Numerical checks and asymptotic meaning

Analytical lines and independent Qualtran call-graph evaluations coincide at every tested size. The multiscale and preconditioned curves almost overlap because their difference is only 4(n-2) Toffolis.



n_ref = 2n + 4. No fitted constants. The two P circuits add only 4(n - 2) Toffolis.

n	Reference bits	LCU	Multiscale	Preconditioned
3	10	1,294	1,346	1,350
4	12	2,136	2,282	2,290
8	20	7,504	8,606	8,630
12	28	16,072	19,442	19,482
16	36	27,840	35,302	35,358

$$\frac{T_{\mathrm{PMS}}}{n^3} = \frac{4}{3} + \frac{109}{n} + \frac{383}{3n^2} - \frac{50}{n^3} \longrightarrow \frac{4}{3}$$

The large $109n^2$ contribution means the leading cubic term alone is not a good numerical estimate at $n \leq 16$. It is not an upper bound. For this fixed precision schedule, the structural formula establishes asymptotic cubic growth; the plot is a finite-size consistency check, not the proof.

Operator validation used classical arithmetic and SELECT actions plus compositional block action at N=8 and 16. It did not simulate or contract the entire quantum circuit. The Toffoli-count agreement is a separate resource-accounting check.

TECHNICAL NOTE / 07

Rotations, scope and traceability

Phase-gradient gates are not phase-gradient kickback

The QFT uses Qualtran's **PhaseGradientUnitary**. In this implementation it decomposes into controlled phase gates with binary-scaled angles. No PhaseGradientState is prepared and no AddIntoPhaseGrad kickback circuit is used. PREP uses generic R_y angles; P uses generic R_z angles. The control-sharing optimization does not make rotation synthesis free.

Rotation audit at n=16, r=36	Count
Generic-angle Ry/Rz	122
Phase rotations still requiring synthesis	1,638
Total unsynthesized rotations	**1,760**
Fixed pi/4-grid rotations (mainly controlled-H decompositions)	1,160
Earlier logical-rotation bucket	2,920

Why these are not the paper's optimal arithmetic constants

The paper estimates an optimized square q^2-q, product 4qr-r and forward comparator 2q+r. Our installed Square and Product bloqs provide resource models but no executable decomposition. We therefore used executable schoolbook products from native controlled adders. The formulas in this note describe that choice, including uncomputation, under the stated coherent-And convention. They must not be presented as the paper's smallest possible constants or compared to a forward-only estimate without reconciling the scope and gate basis.

Finite precision and normalization

$$\alpha_n = \frac{\pi^2}{3} + 8(1 - 2^{-(n-1)}) \longrightarrow \frac{\pi^2 + 24}{3}$$

The initial branch rotation was adjusted to this finite-size normalization so the off-diagonal weights are not spuriously amplified by $1/(1-2^{-(n-1)})$. At finite M the coefficients use the exact ceiling acceptance counts; they only approach $1/j^2$ as M grows. The chosen r=2n+4 is a documented accuracy setting, not a claim that it is minimal. The truncated target is not the exact periodic spectral Laplacian and need not have an exact constant zero mode.

Implementation sources

main.tex: slaccf, the inequality-test state-preparation section, and Appendix SIGN/COPY/SELECT. The paper file was not modified.

slac_multiscale_resources.py: SharedControlQFT, SharedControlQSWTStep, PrefixMultiscaleShannonQSWT and the coherent leaf-cost convention.

slac_complete_inverse_square.py: executable arithmetic, clean inequality oracle and complete LCU composition.

slac_wavelet_preconditioner.py: WaveletScalePreconditioner and shared-prefix controlled rotations.

slac_theoretical_resources.py and **slac_three_circuit_comparison.py:** formula checks, rotation audit and plotted data.

Notebook: slac_qualtran_multiscale_run_order_corrected.ipynb, final sections in cells 43-48. Installed Qualtran addition, comparison and phase_gradient implementations were inspected for the component derivations.